



Curcumin inhibits microglia inflammation and confers neuroprotection in intracerebral hemorrhage



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ARTICLE INFO

Article history:

Received 2 November 2013

Received in revised form 24 February 2014

Accepted 17 March 2014

Available online 25 March 2014

Keywords:

Curcumin

Microglia

Intracerebral hemorrhage

ABSTRACT

Much evidence demonstrates that microglia mediated neuroinflammation is an important contributor to the inflammatory injury in intracerebral hemorrhage (ICH). Therefore, the compounds that can inhibit neuroinflammation are greatly needed. In the current study, we examined whether curcumin, present in a Chinese medicinal plant, could prevent ICH induced microglia activation and confer protection against neurotoxicity.

The cytokines of microglia were measured by ELISA, p38MAPK/PKC and NF- κ B were measured by Western blot and EMSA. Microglial toxicity was assessed using MTT and FACS assays. And neurological function was evaluated by animal behavioristics. We found that curcumin prevented ICH-induced inflammatory molecules through NF- κ B activation via the p38MAPK/PKC pathway in vitro. In addition, curcumin protected hippocampal HT22 cells from indirect toxicity mediated by ICH-treated microglia cells. Further, curcumin also attenuated ICH-induced neurological deficit and cerebral water content in vivo. Together, our findings suggest that curcumin could suppress ICH induced inflammatory injury and represent a novel herbal sources for ICH therapeutical strategy.

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1. Introduction

Intracerebral hemorrhage (ICH), estimated to affect over 1 million people worldwide each year, is the least treatable form of stroke and contributes substantially to the burden of cerebrovascular disease [1–3]. After ICH, a lot of changes occur in the brain, including hematoma formation, brain edema, inflammation and microglia activation [4–6]. Thus, the inhibition of these proinflammatory mediators offers a potentially effective therapeutic approach for ICH.

Curcumin is a hydrophobic polyphenol derived from the rhizomatous herbaceous species *Curcuma longa*, which belongs to the Zingiberaceae family [7–9]. Curcumin exhibits a wide variety of antioxidant, anti-inflammatory, antimicrobial, and anticarcinogenic activities [9–11]. However, the effect of curcumin in ICH has not been well studied.

Therefore, to identify the effect of curcumin in ICH, we test its potential to inhibit inflammatory response in vitro and in vivo.

Our findings demonstrate that curcumin could inhibit microglia inflammation and confer neuroprotection in ICH.

2. Materials and methods

2.1. Cells and animals

The murine microglial cell line BV2 and hippocampal HT22 cells were purchased from the American Type Culture Collection (Manassas, VA) and cultured in Dulbecco's Modified Eagle Medium (DMEM)/F12 supplemented with 10% fetal bovine serum (FBS) (Thermo Scientific Hyclone, Logan, UT, USA), 2 mM glutamine, 100 U/ml penicillin and 100 mg/ml streptomycin. All cell lines mentioned previously were kept at 37°C in a humidified atmosphere containing 5% CO₂. Male C57BL/6 mice (8–10 weeks) were obtained from the Animal Center of the Third Military Medical University and bred under specific pathogen-free conditions. Experiments were conducted in accordance with animal care guidelines approved by the Animal Ethics Committee of the Third Military Medical University. Curcumin (1,7-bis[4-hydroxy-3-methoxyphenyl]-1,6-heptadiene-3,5-dione) and MTT were purchased from Sigma–Aldrich Chemical Co. (St. Louis, MO).

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2.2. Mouse model of intracerebral hemorrhage

Mice were anesthetized with intraperitoneal chloral hydrate (40 mg/kg) and placed in a stereotaxic frame (Stoelting, Kiel, WI, USA). A micro-sample instrument was lowered into the center of the striatum by craniotomy under stereotactic guidance at the following coordinates relative to bregma: 1 mm anterior, 25 mm lateral, and 4 mm deep. A volume of 25 μ L of autologous whole blood was infused at 25 μ L/min over a period of 10 min. The needle was held in place for another 10 min after the infusion to prevent leakage. Craniotomy was then sealed with bone wax, and the scalp was closed with suture. Control mice were infused with 25 μ L of 0.9% saline. We maintained the room temperature at about 25 °C during and after surgery, and exposed the animals to incandescent lighting to keep their rectal temperature at 37 ± 1 °C until palines-thesia. Curcumin was given intraperitoneally as a solution of DMSO, at a dose of 100 mg/kg daily for three days.

2.3. Cell culture

BV2 microglia (1×10^5) were pretreated for 60 min with curcumin (0, 5, 10, and 20 μ M, respectively), and then were stimulated with 10 μ L erythrocyte lysis. After three days, the supernatants were removed and analyzed for cytokine production with ELISA. Hippocampal HT22 cells were cultured in a 96-well plate with 1×10^4 cells per well. For the toxicity experiments, HT22 cells were serum-starved for 4 h and then treated with a mixture of BV2-conditioned medium. For MTT and apoptosis assays, cells were treated for 48 h.

2.4. Enzyme-linked immunosorbent assay (ELISA)

ELISA was performed as per the manufacturer's instructions (Dakewe Biotech, Shenzhen, China) to assess the concentrations of TNF- α , IL-1 β and IL-6 in the culture supernatant.

2.5. Western blot analysis

Briefly, proteins were separated from perihematoma tissues (80 mg) or microglia (1×10^5) by SDS polyacrylamide gel electrophoresis and transferred onto polyvinylidene fluoride (PVDF) membranes (Amersham Pharmacia). The PVDF membranes were incubated with the primary antibodies, including: rabbit anti-mouse p38MAPK (Abcam, UK), rabbit anti-mouse PKC (Abcam, UK), followed by incubation with peroxidase-conjugated secondary antibodies (1:2000, Jingmei, China). The signals were detected with an ECL system (Amersham Pharmacia). The same membranes were probed with antibody for glyceraldehyde-3-phosphate dehydrogenase (GAPDH) after being washed with stripping buffer. The signals were quantified by scanning densitometry and computer-assisted image analysis.

2.6. Determination of NF- κ B activity by electrophoretic mobility shift assay

Nucleoprotein was extracted from brain tissue according to the manufacturer's instructions. Ten microgram of nucleoprotein of each sample was incubated with the reaction buffer at room temperature for 15 min, then P32-labeled oligonucleotide (5'-GGGGACTTCC-3'; Life Technologies, Gaithersburg, MD, USA), which binds to NF- κ B, was added to the reaction buffer and incubated for 15 min. After incubation for 20 min at 25 °C, the reaction mixture was subjected to 6% non-denaturing polyacrylamide gel electrophoresis. Autoradiography was performed at room temperature. Finally, the images were analyzed using an Bio-Rad Image

Analyzer (CA, USA) and the results were expressed as optical density (OD).

2.7. MTT assay

Cell viability of HT22 cells was assessed using 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT, Sigma–Aldrich) assay. After 48 h, MTT reagent was added to the wells, incubated for 4 h at 37 °C, 5% CO₂. After centrifugation, the supernatant was removed from each well. The colored formazan crystal produced from MTT was dissolved with 0.15 mL DMSO, then the optical density (OD) value A490 was measured by the multiscanner autoreader (Dynatech MR 5000; Dynatech Laboratories, Chantilly, VA, USA). The absorbance was measured at 570 nm. The mean of readings of triplicate wells was taken as one value. The OD value for the control cultures was considered as 100% viability and viability in other samples is expressed as a percentage of viability in the control cultures.

2.8. Flow cytometric analysis of apoptosis

An Annexin V-fluorescein isothiocyanate (FITC) apoptosis detection kit (Oncogene Research Products, Boston, MA) was used to detect apoptosis. The cells were seeded in 100 ml flasks and incubated until there was about 90% confluence in DMEM supplemented with 10% fetal bovine serum. Then the cells were harvested, washed with ice-cold PBS twice, and resuspended in binding buffer (10 mM of Hepes, pH 7.4, 150 mM of NaCl, 25 mM of CaCl₂, 1 mM of MgCl₂, 4% bovine serum albumin). Annexin V-fluorescein isothiocyanate (0.5 mg/mL) and propidium iodide (0.6 mg/mL) were then added to a 250 ml aliquot (1×10^6 cells) of this cell suspension according to the protocol of the manufacturer. After a 15 min incubation in the dark at room temperature, stained cells were immediately analyzed on a flow cytometer (Beckman Coulter, USA). All of the samples were assayed in triplicate, and the cell apoptosis rate calculated using the following formula: apoptosis rate = (apoptotic cell number/total cell number) $\times$ 100%.

2.9. Caspase assay

To detect caspase-3-like protease activities, the ApoAlert caspase-3 colorimetric assay kit (Clontech, Palo Alto, USA) was used. Cytosolic lysates were prepared 48 h following transfection and incubated with 50 mM p-nitroanilide (pNA) conjugated to the caspase cleavage site Asp-Glu-Val-Asp (DEVD) for 1 h at 37 °C. Hydrolyzed pNA was detected using a Multiscan MS colorimeter (Thermo Labsystems, Vantaa, Finland) at 405 nm. For control experiments, the lysates were incubated with 10 mM of the caspase-3 inhibitor DEVD-fmk (Clontech) for 30 min, before addition of the substrate.

2.10. Scratch assay

5×10^5 BV2 cells were grown in 6-well plates as 80% confluent monolayers and were wounded with a sterile 100 μ L pipette tip. Thereafter, the cells were stimulated with 10 μ L erythrocyte lysis, 20 μ M of curcumin, 10 μ L erythrocyte lysis + 20 μ M of curcumin, or DMSO as solvent control. Migration into the open scar was documented with microphotographs at different time points after wounding. The number of migrating cells was quantified by counting all cells within a 0.4 mm² region in the center of each scratch. A minimum of five individual cultures was used to calculate the mean migratory capacity of each cell culture condition.

2.11. Transwell migration assay

The Costar Transwell System (8- μ m pore size polycarbonate membrane) was used to evaluate vertical cell migration. 1 Mio BV2 cells in 15 ml serum-free medium were added to the upper well, and 26 ml serum-free medium was added to the lower chamber. 10 μ l erythrocyte lysis, 20 μ m curcumin, 10 μ l erythrocyte lysis + 20 μ m curcumin, or DMSO as solvent control were added to the lower chamber medium. At the end of a 24 h incubation period, cells that had migrated to the lower surface were quantified by counting the migrated cells on the lower surface of the membrane using microscopy.

2.12. Neurological deficit scores

After 3-day post ICH modeling, the mice were in stable condition, the neurological deficit tests were performed by behavioral measurement, including postural flexing test, circling or sidewalk, forelimb placing and foot fault test and repeated three times.

2.13. Measurement of cerebral water content of mice

To measure cerebral water content after ICH, mice were randomly selected from each group and euthanized. We levered the skull within 1 min to take out brain tissues, blotted up the water on the surface of the left hemisphere with filter paper, and took the humid weights (GW) on an electronic balance. We then dried them for 24–48 h at 95–100 °C in an Electro-Thermostatic Blast Oven and took their dry weights (DW). Cerebral water contents were calculated by the formula: cerebral water content% = (GW – DW)/GW $\times$ 100%.

2.14. Statistical analysis

The statistical significance of differential findings between experimental groups and controls were determined by *t*-test and considered significant if *P* < 0.05.

3. Results

3.1. Curcumin decreased inflammatory cytokine in ICH induced BV2 microglia

To evaluate the anti-inflammatory effects of curcumin in ICH induced BV2 microglia, cell culture media were collected, and TNF- α , IL-1 β and IL-6 levels were measured in ICH induced BV2 microglia. As Fig. 1 demonstrated that the amount of TNF- α , IL-1 β and IL-6 significantly increased 24 h after erythrocyte lysis stimulation, curcumin suppressed the related inflammatory cytokine at a concentration of 10 and 20 μ mol/l, respectively. These results indicated that curcumin significantly inhibited inflammatory cytokine production in ICH induced BV2 microglia in a dose-dependent manner.

3.2. Curcumin inhibited microglia activation through p38MAPK/PKC pathway and NF- κ B activation

Previous evidence demonstrated that neuronal p38MAPK/PKC signaling and NF- κ B were emerging regulator of cell fate and function in the nervous system. In this study, we also evaluated p38MAPK/PKC and NF- κ B activation 24 h after stimulus with erythrocyte lysis. As shown in Fig. 2, erythrocyte lysis induced a significant increase of p38MAPK/PKC and NF- κ B. However, curcumin could inhibit p38MAPK/PKC and NF- κ B expression. The results demonstrated curcumin inhibited ICH induced

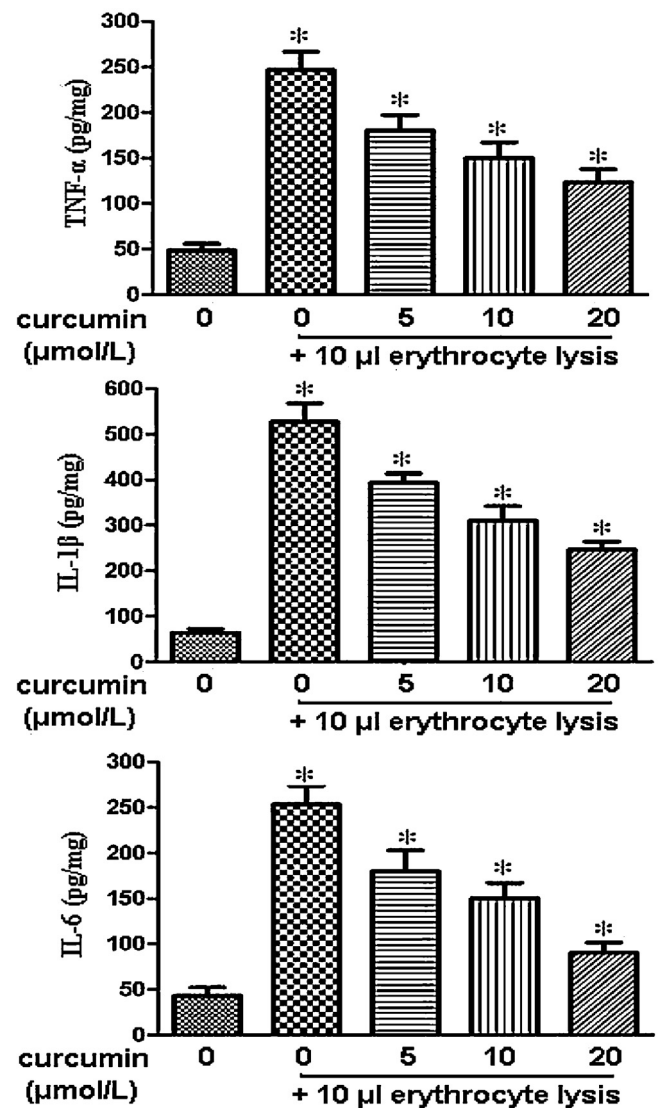


Fig. 1. Detection of cytokine secretion of microglia. BV2 microglia (1×10^5) were pretreated for 60 min with curcumin (0, 5, 10, and 20 μ mol/l, respectively), and then were stimulated with 10 μ l erythrocyte lysis. After three days, the supernatants were removed and analyzed for cytokine production with ELISA. Experiments performed in triplicate showed consistent results. Compared with controls, **P* < 0.05.

microglia activation through p38MAPK/PKC pathway and NF- κ B activation.

3.3. Curcumin protected hippocampal HT22 cells against toxicity

The activated microglia can cause toxicity to neurons. We next explored whether curcumin could inhibit toxicity of activated microglia to hippocampal cells. We used an MTT assay to examine the effect of curcumin on erythrocyte lysis-induced toxicity to HT22 cells. We found that when HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells, there was a significant decrease in cell viability. However, when the cells were treated with conditioned medium from BV2 cells treated with erythrocyte lysis and curcumin, the cell viability was increased (as shown in Fig. 3A). Thus, curcumin protected HT22 cells against toxicity induced by ICH.

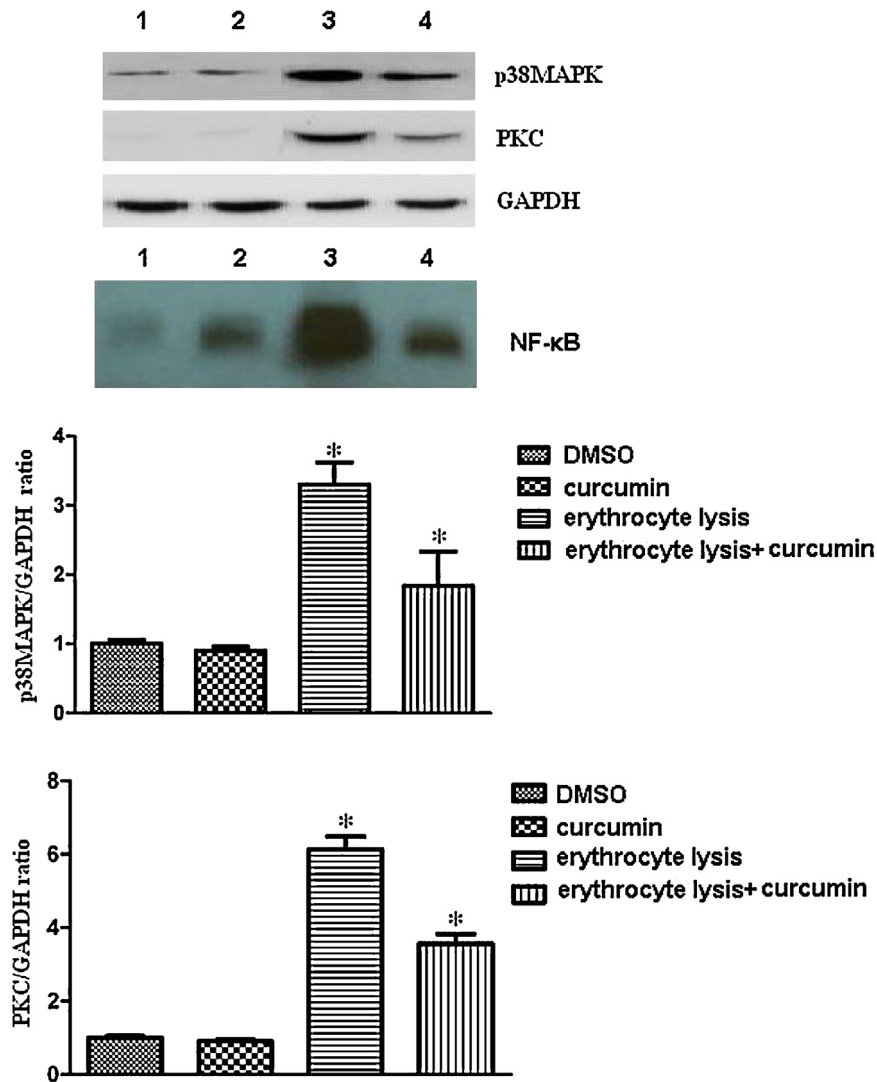


Fig. 2. Analysis of p38MAPK/PKC and NF-κB activity of microglia. BV2 microglia (1×10^5) were pretreated for 60 min with 20 μ M/L curcumin, and then were stimulated with 10 μ l erythrocyte lysis. After three days, p38MAPK/PKC and NF-κB activity of microglia were analyzed. (1) DMSO; (2) curcumin; (3) erythrocyte lysis; and (4) curcumin + erythrocyte lysis. Experiments performed in triplicate showed consistent results. Compared with controls, * $P < 0.05$.

3.4. Curcumin inhibited hippocampal HT22 cells apoptosis

To detect whether curcumin could inhibit apoptosis of HT22 cells, two days after HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells, cell apoptosis ratio was detected by flow cytometry. Apoptosis cells were determined by Annexin V positive and propidium iodide (PI) negative cells. The results demonstrated that when HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells, there was a significant increase in cell apoptosis rate. However, when the cells were treated with conditioned medium from BV2 cells treated with erythrocyte lysis and curcumin, the cell apoptosis rate was decreased (as shown in Fig. 3B). Thus, curcumin protected HT22 cells against apoptosis induced by ICH.

3.5. Effect of curcumin on activation of apoptotic pathways

To determine the activation of caspase during ICH-induced apoptosis of HT22 cells, we investigated caspase-specific activities after HT22 cells were treated with conditioned medium from

erythrocyte lysis-treated BV2 cells. As shown in Fig. 3C, the colorimetric assay of caspase activity revealed that the caspase-3 activity of the HT22 cells group remarkably increased compared with control group treated with conditioned medium from BV2 cells treated with erythrocyte lysis and curcumin.

3.6. Curcumin inhibited microglia migration

To explore the effect of curcumin on microglia migration, we cultured BV2 microglia on plastic dishes until 80% confluence and then created a scratch with a pipette tip. 12 h after stimulation of resting microglia or erythrocyte lysis-activated cells with 20 μ M of curcumin, migration into the cell-free scratch area was documented. The statistical analysis of five independent experiments revealed a significantly reduced number of migrating cells when curcumin was present in the culture medium (Fig. 4A). As an independent measure of microglia cell motility and to study the long term effects of curcumin, we performed transwell migration assays over a period of 24 h. Similar as in the scratch assays, curcumin caused a significant attenuation of microglia migration (Fig. 4B).

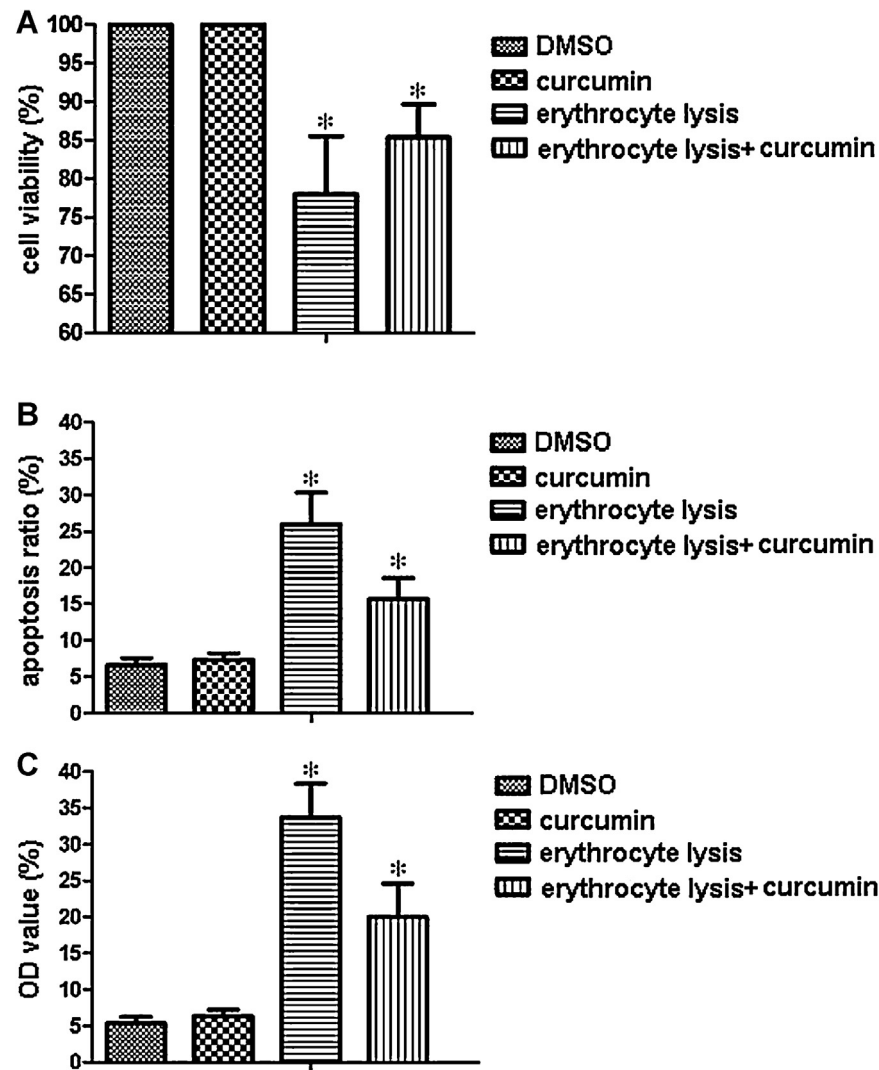


Fig. 3. Curcumin enhanced cell viability and inhibited cell apoptosis of HT22 cells. (A) HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells. After 48 h, MTT reagent was added and the cell viability was assessed. (B) 48 h after HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells, cell apoptosis ratio was detected by flow cytometry. Apoptosis cells were determined by Annexin V positive and propidium iodide (PI) negative cells. (C) 48 h after HT22 cells were treated with conditioned medium from erythrocyte lysis-treated BV2 cells, the ApoAlert caspase-3 colorimetric assay kit (Clontech, Palo Alto, USA) was used. Hydrolyzed pNA was detected using a Multiscan MS colorimeter (Thermo Labsystems, Vantaa, Finland) at 405 nm. Experiments performed in triplicate showed consistent results. Compared with controls, * $P < 0.05$.

3.7. Curcumin inhibited inflammation and neurological impairment following ICH

To further assess the role of curcumin in the inflammatory response to ICH, we injected curcumin into mice intraperitoneally at a dose of 100 mg/kg daily for three days, and then evaluated brain damage and neurological impairments three days after ICH. The results demonstrated that curcumin could inhibit brain water content and neurological deficit scores (Fig. 5A and B).

3.8. Curcumin inhibited microglia activation in vivo

To identify whether curcumin could inhibit microglia activation in vivo, we also analyzed inflammatory cytokines expression in mice after ICH. The results showed that curcumin could significantly inhibit expression of TNF- α , IL-1 β and IL-6 in perihematoma tissues three days after ICH (Fig. 5C).

4. Discussion

Intracerebral hemorrhage (ICH) is a common type of fatal stroke, accounting for about 15–20% of all strokes [12–14]. Hemorrhagic strokes are associated with high mortality and morbidity, and increasing evidence shows that innate immune responses and inflammatory injury play a critical role in ICH-induced neurological deficits [15–17]. Although there is much research on the events following ICH, the initial cellular events trigger innate immune and inflammatory responses are still complex. Therefore, the strategy for inhibiting the inflammatory responses is greatly needed.

Curcumin is a major chemical component of turmeric (*Curcuma longa*) which is used as a spice to give a specific flavor and yellow color in curry [18,19]. The form of herbal powder turmeric has been used for centuries as an anti-inflammatory remedy in Asian medicine. Curcumin has been shown to display anti-inflammatory, antioxidant, and anticarcinogenic properties [20–22]. The anti-inflammatory effects of curcumin are most likely mediated through

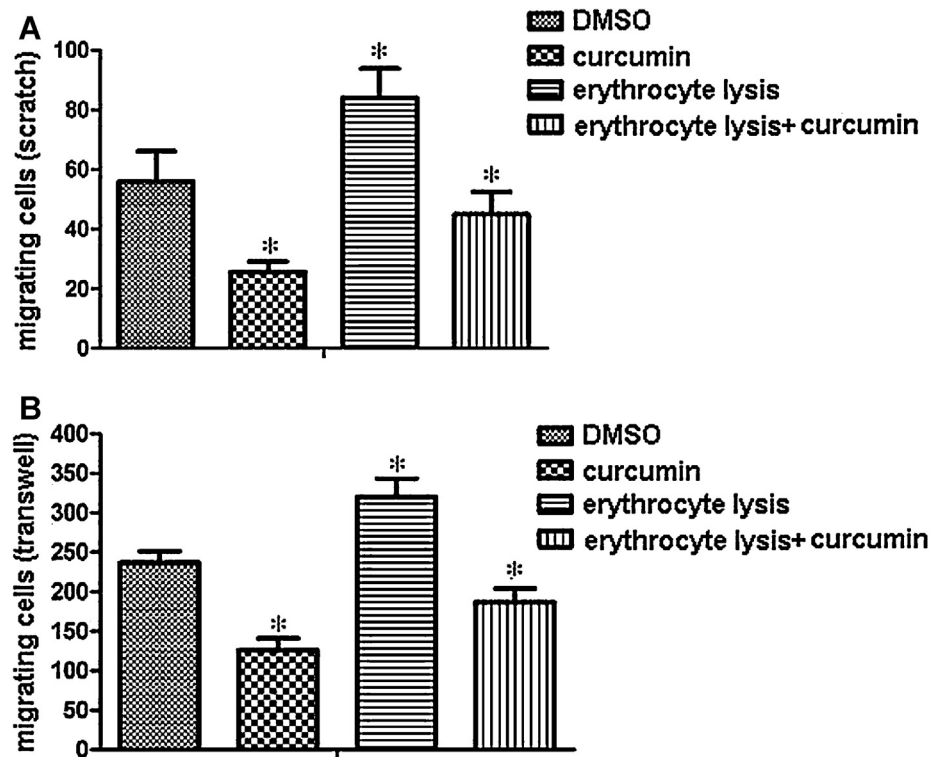


Fig. 4. Curcumin reduced microglial migration. (A) 5×10^5 BV-2 cells were grown in 6-well plates as 80% confluent monolayers and were wounded with a sterile 100 μ l pipette tip. Thereafter, the cells were stimulated with 10 μ l erythrocyte lysis, 20 μ M of curcumin, 10 μ l erythrocyte lysis + 20 μ M of curcumin, or DMSO as solvent control. Migration into the open scar was documented with microphotographs at different time points after wounding. The number of migrating cells was quantified by counting all cells within a 0.4 mm² region in the center of each scratch. (B) 1 Mio BV-2 cells in 15 ml serum-free medium were added to the upper well, and 26 ml serum-free medium was added to the lower chamber. 10 μ l erythrocyte lysis, 20 μ M curcumin, 10 μ l erythrocyte lysis + 20 μ M curcumin, or DMSO as solvent control were added to the lower chamber medium. At the end of a 24 h incubation period, cells that had migrated to the lower surface were quantified by counting the migrated cells on the lower surface of the membrane using microscopy. Experiments performed in triplicate showed consistent results. Compared with controls, * $P < 0.05$.

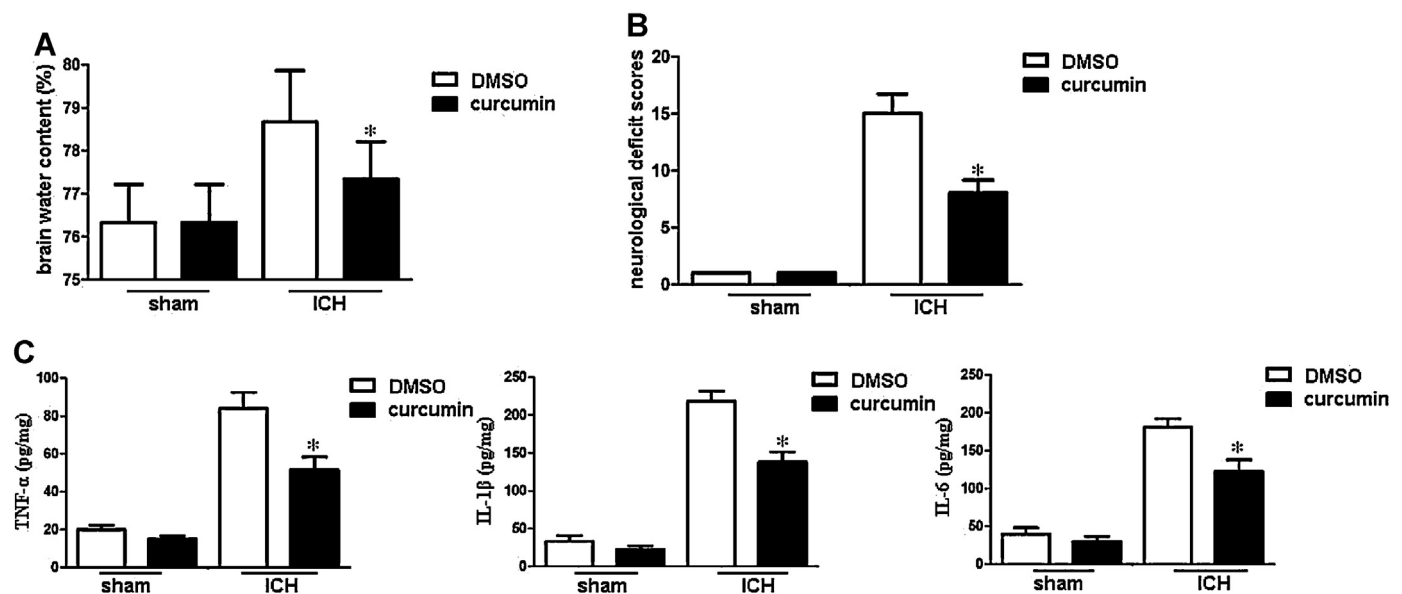


Fig. 5. Analysis of inflammation and neurological impairment following ICH. (A). After 3-day post ICH modeling, the cerebral water content of mice were analyzed. (B). After 3-day post ICH modeling, the mice were in stable condition, the neurological deficit tests were performed by behavioral measurement, including postural flexing test, circling or sidewalk, forelimb placing and foot fault test and repeated three times. (C). After 3-day post ICH modeling, the concentrations of TNF- α , IL-1 β and IL-6 in brain tissues obtained from the perihematoma region were analyzed with ELISA. Brain tissues (80 mg) were centrifuged at 12,000 $\times$ g and the supernatant was collected for analysis. Experiments performed in triplicate showed consistent results. Compared with controls, * $P < 0.05$.

its ability to inhibit expression of pro-inflammatory genes such as cyclooxygenase-2, adhesion molecules, chemokines, cytokines, metalloproteinases (MMP), lipoxygenase, and iNOS [23,24,25]. A recent study has shown that curcumin reduces oxidative damage and amyloid pathology in an Alzheimer transgenic mouse [26]. In addition, curcumin has bright prospects for the treatment of multiple sclerosis through inhibition of IL-17 production [27,28]. Although the anti-inflammatory properties of curcumin are known, its molecular basis in ICH is not fully understood yet.

In the current study, to evaluate the anti-inflammatory effects of curcumin in ICH induced BV2 microglia, we firstly analyzed the cytokines of BV2 microglia 24 h after erythrocyte lysis stimulation. The results indicated that curcumin significantly inhibited inflammatory cytokine production in ICH induced BV2 microglia in a dose-dependent manner. We also explored the anti-inflammatory properties of curcumin, and the results demonstrated that curcumin could inhibit erythrocyte lysis induced p38MAPK/PKC and NF- κ B expression. In addition, to explore the effect of curcumin on microglial migration, we performed the scratch assay and transwell migration assay, and the results demonstrated that curcumin caused a significant attenuation of microglial migration.

The related studies demonstrated that the activated microglia can cause toxicity to neurons. In order to explore whether curcumin could inhibit toxicity of activated microglia to hippocampal cells, we still examined the effect of curcumin on erythrocyte lysis-induced toxicity to HT22 cells and observed the cell apoptosis. We found that when the cells were treated with conditioned medium from BV2 cells treated with erythrocyte lysis and curcumin, the cell viability increased and the cell apoptosis rate decreased though caspase-3 activity.

Lastly, we injected curcumin into mice intraperitoneally in ICH model, to identify whether curcumin could inhibit inflammation and neurological impairment following ICH. We evaluated brain damage and neurological impairments three days after ICH. The results demonstrated that curcumin could inhibit brain water content and neurological deficit scores. Additionally, the results showed that curcumin could significantly inhibited expression of TNF- α , IL-1 β and IL-6 in perihematoma tissues three days after ICH. All the data suggested that curcumin could inhibit microglia activation *in vivo*.

Taken together, our study suggests that curcumin is a promising strategy to inhibit inflammation and neurological impairment in ICH, and might be utilized in the clinic.

Acknowledgments

This study was supported by a grant from National Natural Science Foundation of China (No. 81200908), National Basic Research Program of China (973Program) (No. 2014CB541600) and Major International (Regional) Joint Research Project of China (No. 81220108009).

References

- [1] Florczak-Rzepka M, Grond-Ginsbach C, Montaner J, Steiner T. Matrix metalloproteinases in human spontaneous intracerebral hemorrhage: an update. *Cerebrovasc Dis* 2012;34(4):249–62.
- [2] Lee HK, Ghani AR, Awang MS, Sayuthi S, Idris B, Abdullah JM. Role of high augmentation index in spontaneous intracerebral haemorrhage. *Asian J Surg* 2010;33(1):42–50.
- [3] Anderson CS. Medical management of acute intracerebral hemorrhage. *Curr Opin Crit Care* 2009;15(2):93–8.
- [4] Tanaka Y, Marumo T, Shibuta H, Omura T, Yoshida S. Serum S100B, brain edema, and hematoma formation in a rat model of collagenase-induced hemorrhagic stroke. *Brain Res Bull* 2009;78(4/5):158–63.
- [5] Zhou QB, Jin YL, Jia Q, Zhang Y, Li LY, Liu P, et al. Baicalin attenuates brain edema in a rat model of intracerebral hemorrhage. *Inflammation* 2014;37(1):107–15.
- [6] Wang J. Preclinical and clinical research on inflammation after intracerebral hemorrhage. *Prog Neurobiol* 2010;92(4):463–77.
- [7] Guri A, Gulseren I, Corredig M. Utilization of solid lipid nanoparticles for enhanced delivery of curcumin in cocultures of HT29-MTX and Caco-2 cells. *Food Funct* 2013;4(9):1410–9.
- [8] Liu J, Chen S, Lv L, Song L, Guo S, Huang S. Recent progress in studying curcumin and its nano-preparations for cancer therapy. *Curr Pharm Des* 2013;19(11):1974–93.
- [9] Chen CC, Chan WH. Injurious effects of curcumin on maturation of mouse oocytes, fertilization and fetal development via apoptosis. *Int J Mol Sci* 2012;13(4):4655–72.
- [10] Bar-Sela G, Epelbaum R, Schaffer M. Curcumin as an anti-cancer agent: review of the gap between basic and clinical applications. *Curr Med Chem* 2010;17(3):190–7.
- [11] Altunbas A, Lee SJ, Rajasekaran SA, Schneider JP, Pochan DJ. Encapsulation of curcumin in self-assembling peptide hydrogels as injectable drug delivery vehicles. *Biomaterials* 2011;32(25):5906–14.
- [12] Chen J, Qin J, Su Q, Liu Z, Yang J. Treadmill rehabilitation treatment enhanced BDNF-TrkB but not NGF-TrkA signaling in a mouse intracerebral hemorrhage model. *Neurosci Lett* 2012;529(1):28–32.
- [13] Haussen DC, Henninger N, Kumar S, Selim M. Statin use and microbleeds in patients with spontaneous intracerebral hemorrhage. *Stroke* 2012;43(10):2677–81.
- [14] Cordonnier C, Klijn CJ, van Beijnum J, Al-Shahi Salman R. Radiological investigation of spontaneous intracerebral hemorrhage: systematic review and trinitational survey. *Stroke* 2010;41(4):685–90.
- [15] Lee ST, Chu K, Jung KH, Kang KM, Kim JH, Bahn JJ, et al. Cholinergic anti-inflammatory pathway in intracerebral hemorrhage. *Brain Res* 2010;1309:164–71.
- [16] Munakata M, Shirakawa H, Nagayasu K, Miyahara J, Miyake T, Nakagawa T, et al. Transient receptor potential canonical 3 inhibitor Pyr3 improves outcomes and attenuates astroglial activation after intracerebral hemorrhage in mice. *Stroke* 2013;44(7):1981–7.
- [17] Tang XN, Zheng Z, Giffard RG, Yenari MA. Significance of marrow-derived nicotinamide adenine dinucleotide phosphate oxidase in experimental ischemic stroke. *Ann Neurol* 2011;70(4):606–15.
- [18] Aggarwal BB, Yuan W, Li S, Gupta SC. Curcumin-free turmeric exhibits anti-inflammatory and anticancer activities: Identification of novel components of turmeric. *Mol Nutr Food Res* 2013;57(9):1529–42.
- [19] Cine N, Limtrakul P, Sunnetci D, Nagy B, Savli H. Effects of curcumin on global gene expression profiles in the highly invasive human breast carcinoma cell line MDA-MB 231: a gene network-based microarray analysis. *Exp Ther Med* 2013;5(1):23–7.
- [20] Kloesch B, Becker T, Dietersdorfer E, Kiener H, Steiner G. Anti-inflammatory and apoptotic effects of the polyphenol curcumin on human fibroblast-like synoviocytes. *Int Immunopharmacol* 2013;15(2):400–5.
- [21] Singh DK, Jagannathan R, Khandelwal P, Abraham PM, Poddar P. In situ synthesis and surface functionalization of gold nanoparticles with curcumin and their antioxidant properties: an experimental and density functional theory investigation. *Nanoscale* 2013;5(5):1882–93.
- [22] Zlotogorski A, Dayan A, Dayan D, Chaushu G, Salo T, Vered M. Nutraceuticals as new treatment approaches for oral cancer: I. Curcumin. *Oral Oncol* 2013;49(3):187–91.
- [23] Moon DO, Kim MO, Choi YH, Park YM, Kim GY. Curcumin attenuates inflammatory response in IL-1 β -induced human synovial fibroblasts and collagen-induced arthritis in mouse model. *Int Immunopharmacol* 2010;10(5):605–10.
- [24] Meng Z, Yan C, Deng Q, Gao DF, Niu XL. Curcumin inhibits LPS-induced inflammation in rat vascular smooth muscle cells *in vitro* via ROS-related TLR4-MAPK/NF- κ B pathways. *Acta Pharmacol Sin* 2013;34(7):901–11.
- [25] Ben P, Liu J, Lu C, Xu Y, Xin Y, Fu J, et al. Curcumin promotes degradation of inducible nitric oxide synthase and suppresses its enzyme activity in RAW 264.7 cells. *Int Immunopharmacol* 2011;11(2):179–86.
- [26] Lim GP, Chu T, Yang F, Beech W, Frautschi SA, Cole GM. The curry spice curcumin reduces oxidative damage and amyloid pathology in an Alzheimer transgenic mouse. *J Neurosci* 2001;21(21):8370–7.
- [27] Xie L, Li XK, Takahara S. Curcumin has bright prospects for the treatment of multiple sclerosis. *Int Immunopharmacol* 2011;11(3):323–30.
- [28] Xie L, Li XK, Funeshima-Fuji N, Kimura H, Matsumoto Y, Isaka Y, et al. Amelioration of experimental autoimmune encephalomyelitis by curcumin treatment through inhibition of IL-17 production. *Int Immunopharmacol* 2009;9(5):575–81.